Vinay

Firmware / Embedded Software Engineer
VIT Chennai | B.Tech ECE | GPA: 9.14 / 10 | 2026
VLSI Design Certificate -Maven Silicon

Vinaysharma17005@gmail.com
+91-8010471469 | Faridabad, India
/ Vinay-ix | /Vinay-ix
/ Vinaysh.in

Summary

Recently graduated ECE student from VIT Chennai (GPA 9.14) with experience in embedded firmware, IoT systems, FPGA design, and real-time communication protocols (UART/SPI/I2C/CAN/MQTT). Skilled in Embedded C/C++, Python, FreeRTOS, and hardware debugging. Seeking Embedded Software, Firmware roles

Skills & Tools

Programming & Scripting	: C, C++, Embedded C, Python
Embedded & Real-Time Systems	: FreeRTOS, Interrupt Handling, DMA, PWM, ADC/DAC, Sensor Integration
Communication Protocols	: UART, SPI, I²C, CAN, MQTT
Embedded Platforms	: ESP32, Arduino Uno, Jetson Nano .
Digital Design & Hardware	: SystemVerilog, FPGA Design , RTL Design .
Tools & Debugging	: Git, Oscilloscope, Logic Analyzer ,ModelSim, Intel Quartus Prime
Soft Skills	: Problem Solving , Team Collaboration ,Troubleshooting, Critical Thinking

Internship Experience

JKW Innovatics Pvt. Ltd. — Graduate Engineer Trainee — Greater Noida, Uttar Pradesh | Jul 2026 – Present

- Developing ROS2-based control and navigation systems for autonomous warehouse AMR/AVR robots.
- Working on LiDAR- and IMU-based localization, odometry, and EKF sensor fusion to improve positioning accuracy and stability.
- Developing motion-control and path-tracking algorithms for reliable autonomous robot navigation in warehouse environments.
- Testing, debugging, and tuning autonomous navigation systems using ROS2, Gazebo, RViz, AMCL, LiDAR, and IMU under real-world operating conditions.

Mafkin Robotics Pvt. Ltd. — *ECE Intern* — Chennai, TN | Oct 2024 – Jan 2025

- `Debugged and integrated embedded subsystems (microcontroller firmware, sensors, communication interfaces) for underwater ROV platforms using structured root-cause analysis and oscilloscope / logic-analyzer testing, reducing issue-resolution time by
  ~25% and integration-test failures by ~20%.
- Collaborated across hardware and software teams to integrate sensor modules and control logic, improving system response time and overall reliability.
- Gained direct exposure to UART/SPI debug workflows and real-world firmware validation in a cross-functional engineering environment.

Projects

AI Power Touch-Free Alert System | *ESP32 · Jetson Nano · MQTT · OpenCV · Python* — Apr 2026

- Integrated ESP32-based sensor nodes with MQTT communication and OpenCV-based blink detection on Jetson Nano to deliver real-time patient alerts with end-to-end latency under 2 seconds.
- Designed low-latency MQTT pipeline from ESP32 nodes to a centralized dashboard; added Telegram Bot API integration for automated emergency alerts.
- Demonstrated multi-platform embedded + edge-AI integration — MCU firmware, broker-based messaging, and computer-vision inference running concurrently.

Real-Time Avionics Primary Flight Display | *C++ · Python · UART · Sensor Data Processing* — Sep 2025

- Built a real-time data acquisition and visualization pipeline for sensor-based flight parameters including noise-filtering algorithms to stabilize readings.
- Implemented UART-based serial data transmission between sensor nodes and display; resolved latency and data-consistency issues to optimize end-to-end throughput.
- Outcome: stable, low-jitter display refresh suitable for primary flight instrument simulation.

Real-Time Lossless JPEG-LS Compression on FPGA | *SystemVerilog · Intel Quartus Prime · ModelSim* — May 2025

- Implemented and validated a high-throughput FPGA image compression pipeline; fixed timing violations and clock-domain-crossing issues to reach stable 103 MHz operation — a 4× throughput gain over initial implementation.
- Verified full correctness via ModelSim simulation and Quartus timing analysis.

Education

Vellore Institute of Technology — *B.Tech – Electronics and Communication Engineering* Chennai, TN | 2022 – 2026

GPA: 9.14 / 10 | Relevant coursework: Embedded Systems, Digital Design, IoT, Signals & Communication

St. Joseph International School — *Higher Secondary (MPC)* Kota, RJ | 2019 – 2020

Percentage : 87/100 %

Modern Arya Public School — *Secondary School (Science)* Faridabad, HR | 2017 – 2018

Percentage : 83.4/100 %

Certifications & Achievements

- Digital Design Training – Maven Silicon — *Verilog, RV32I RISC-V processor fundamentals, UVM verification basics, AHB interface design (Jun–Jul 2024)*
- GATE (ECE) Qualified
- 300+ DSA problems solved on LeetCode
- Coordinated finance awareness session – Dream Merchants Club, VIT Chennai — *event logistics and audience engagement*